



# AADHAAR ENROLMENT ANALYTICS

**Aadhaar Enrolment Demand & Resource Planning**

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**UIDAI HACKATHON 2026**

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## **PROBLEM STATEMENT :**

*“Understanding Aadhar Enrolment Demand Patterns for Optimized Resource Planning.”*

## **APPROACH :**

- Analysed anonymized Aadhar enrolment data across states, months, and age groups.
- Cleaned and aggregated daily data into monthly summaries.
- Performed descriptive analysis to identify trends, spikes and regional variations.
- Build interactive Power BI dashboard to visualise enrolment patterns.
- Converted analytical findings into insights for demand forecasting and resource planning.

## Datasets Used :

The insights are derived from a comprehensive analysis of official UIDAI Aggregated Enrolment Datasets.

- api\_data\_aadhar\_enrolment\_0\_500000.csv
- api\_data\_aadhar\_enrolment\_500000\_1000000.csv
- api\_data\_aadhar\_enrolment\_1000000\_1006029.csv

Only UIDAI-provided datasets were used, as per hackathon rules.

## METHODOLOGY :

**Raw CSV  
Ingestion**

**Data Cleaning &  
Validation**

**Data  
Transformation**

**Exploratory Data  
Analysis**

**Data  
Visualisation**

**Insight  
Generation**

**Resource Planning  
Implications**



## TOOLS USED :

- Excel/ CSV Preprocessing
- Jupyter Notebook (Data cleaning using Python and libraries)
- Power BI (DAX and Visualisation)

## ANALYSIS and VISUALISATION :

### CSV Ingestion

```
en1 = pd.read_csv(r"C:\Users\INDIA\Desktop\self_project\UIDAI\api_data_aadhar_enrolment\api_data_aadhar_enrolment_0_500000.csv")
en2 = pd.read_csv(r"C:\Users\INDIA\Desktop\self_project\UIDAI\api_data_aadhar_enrolment\api_data_aadhar_enrolment_500000_1000000.csv")
en3 = pd.read_csv(r"C:\Users\INDIA\Desktop\self_project\UIDAI\api_data_aadhar_enrolment\api_data_aadhar_enrolment_1000000_1006029.csv")
df = pd.concat([en1, en2, en3], ignore_index=True)
df.sample(2)
```

|        | date       | state       | district | pincode | age_0_5 | age_5_17 | age_18_greater |
|--------|------------|-------------|----------|---------|---------|----------|----------------|
| 152476 | 11-09-2025 | Maharashtra | Mumbai   | 400089  | 1       | 0        | 0              |
| 429581 | 18-10-2025 | Rajasthan   | Jaipur   | 303104  | 5       | 2        | 0              |

### Simple Analysis ¶

```
[6]: df.shape
```

```
[6]: (1006029, 7)
```

```
[7]: df.duplicated().sum()
```

```
[7]: np.int64(22957)
```

```
[8]: df = df.drop_duplicates()
df.shape
```

```
[8]: (983072, 7)
```

```
[9]: df.isnull().sum()
```

```
[9]: date      0
state      0
district    0
pincode     0
age_0_5     0
age_5_17    0
age_18_greater  0
dtype: int64
```

## ▼ Unique States

```
[10]: len(df['state'].unique())
```

```
[10]: 55
```

```
[11]: # Step 1: normalize again (safe even if already done)
```

```
df['state'] = (  
    df['state']  
    .astype(str)  
    .str.lower()  
    .str.strip()  
    .str.replace(r'^a-z\s', '', regex=True)  
    .str.replace(r'\s+', ' ', regex=True)  
)
```

```
# Step 2: merge ALL known variants into correct forms
```

```
df['state'] = df['state'].replace({
```

```
    # West Bengal
```

```
    'west bangal': 'west bengal',
```

```
    # Jammu & Kashmir
```

```
    'jammu kashmir': 'jammu and kashmir',
```

```
    # Andaman & Nicobar
```

```
    'andaman nicobar islands': 'andaman and nicobar islands',
```

```
    # Dadra & Daman combinations
```

```
    'dadra nagar haveli': 'dadra and nagar haveli and daman and diu',
```

```
    'dadra and nagar haveli': 'dadra and nagar haveli and daman and diu',
```

```
    'daman and diu': 'dadra and nagar haveli and daman and diu',
```

```
    'daman diu': 'dadra and nagar haveli and daman and diu',
```

```
    'the dadra and nagar haveli and daman and diu': 'dadra and nagar haveli and daman and diu',
```

```
    # Remove empty values
```

```
    '': None
```

```
})
```

```
# Step 3: map to OFFICIAL names
```

```
state_mapping = {
```

```
    'andhra pradesh': 'Andhra Pradesh',
```

```
    'arunachal pradesh': 'Arunachal Pradesh',
```

```
    'assam': 'Assam',
```

```
    'bihar': 'Bihar',
```

```
    'chhattisgarh': 'Chhattisgarh',
```

```
    'goa': 'Goa',
```

```
    'gujarat': 'Gujarat',
```

```
    'haryana': 'Haryana',
```

```
    'himachal pradesh': 'Himachal Pradesh',
```

```
    'jharkhand': 'Jharkhand',
```

```
    'karnataka': 'Karnataka',
```

```
    'kerala': 'Kerala',
```

```

'madhya pradesh': 'Madhya Pradesh',
'maharashtra': 'Maharashtra',
'manipur': 'Manipur',
'meghalaya': 'Meghalaya',
'mizoram': 'Mizoram',
'nagaland': 'Nagaland',
'odisha': 'Odisha',
'punjab': 'Punjab',
'rajasthan': 'Rajasthan',
'sikkim': 'Sikkim',
'tamil nadu': 'Tamil Nadu',
'telangana': 'Telangana',
'tripura': 'Tripura',
'uttar pradesh': 'Uttar Pradesh',
'uttarakhand': 'Uttarakhand',
'west bengal': 'West Bengal',

# UTs
'andaman and nicobar islands': 'Andaman and Nicobar Islands',
'chandigarh': 'Chandigarh',
'delhi': 'Delhi',
'jammu and kashmir': 'Jammu and Kashmir',
'ladakh': 'Ladakh',
'lakshadweep': 'Lakshadweep',
'puducherry': 'Puducherry',
'dadra and nagar haveli and daman and diu': 'Dadra and Nagar Haveli and Daman and Diu'
}

```

```

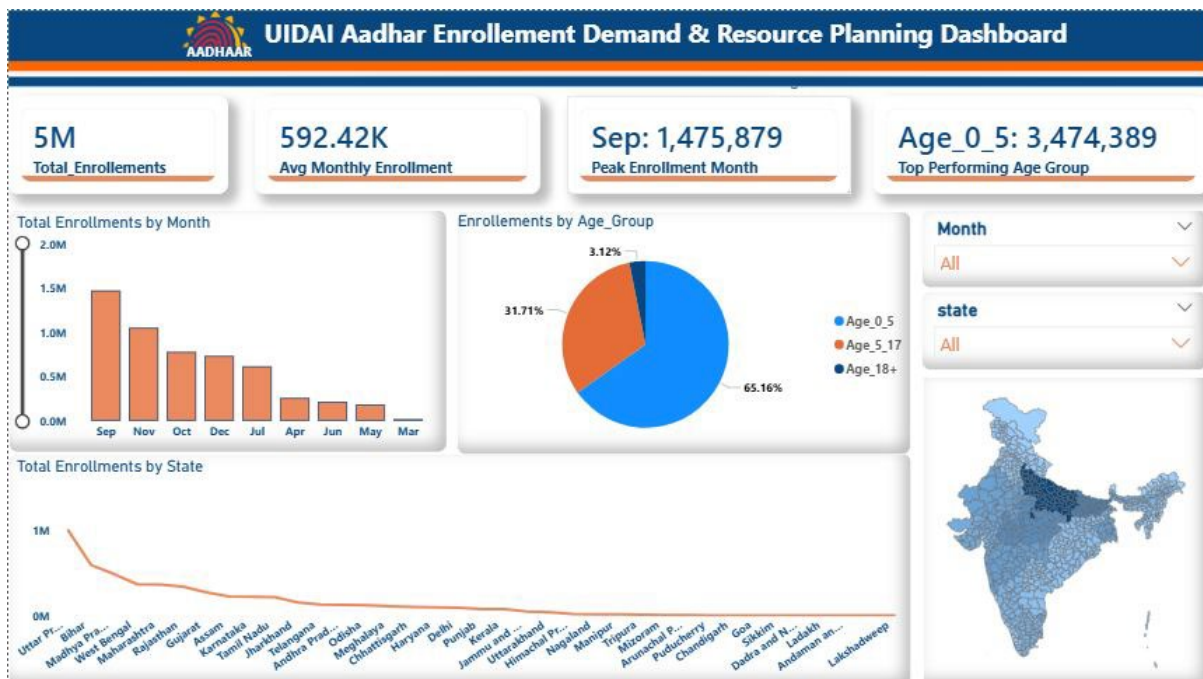
df['state'] = df['state'].map(state_mapping)

# Step 4: final verification
print("Unique states / UTs:", df['state'].nunique())

```

Unique states / UTs: 36

```
[ ]: df.to_csv("uidai_enrolment_cleaned_2.csv", index=False)
```

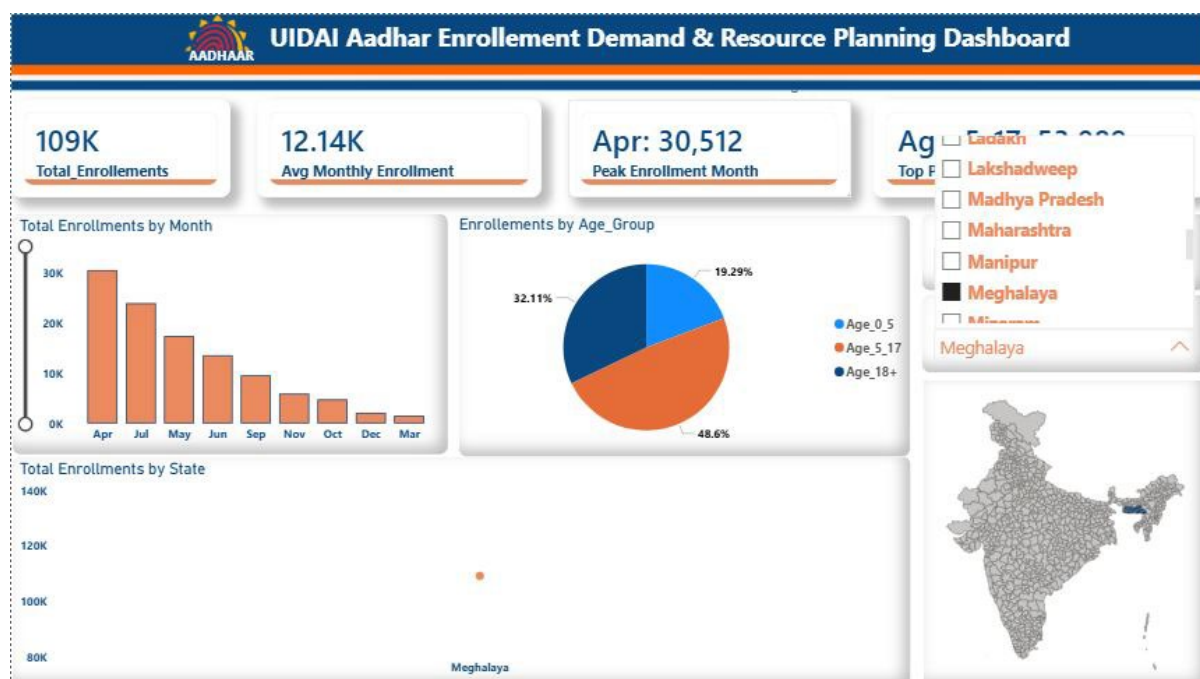


## KEY INSIGHTS & ACTIONABLE IMPLICATIONS :

### ➔ Seasonal Demand Variation :

Most states record peak Aadhar enrolments between **September-December**, whereas **Meghalaya and Mizoram** show peaks during **April-July**.

*Recommendation* : State specific **staff deployment and temporary enrolment centres** should be planned on regional peak months.

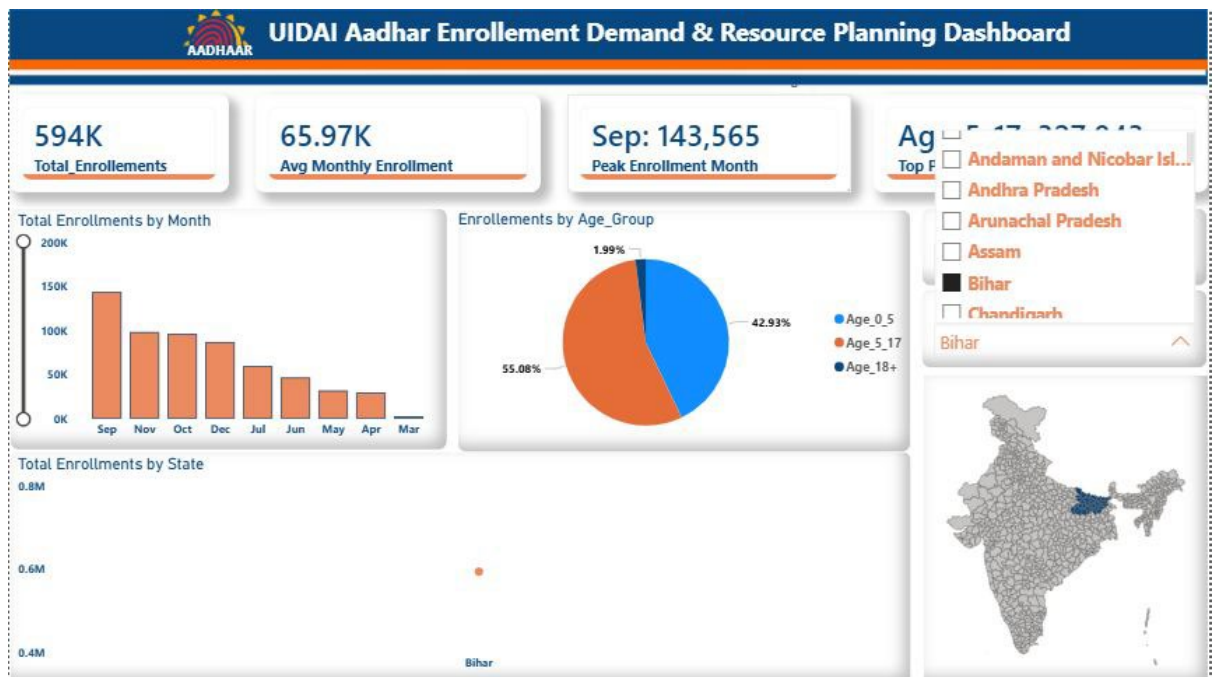


### ➔ Age Group Enrolment Shift :

While the **0-5 age group** dominates enrolments across most states, **Bihar, UP, Manipur, Meghalaya, Nagaland, and Sikkim** show higher enrolment in the **5-17 age group**.

*Recommendation* : Enrolment infrastructure should be strengthened in **schools and adolescent-focused institutions** in these states.

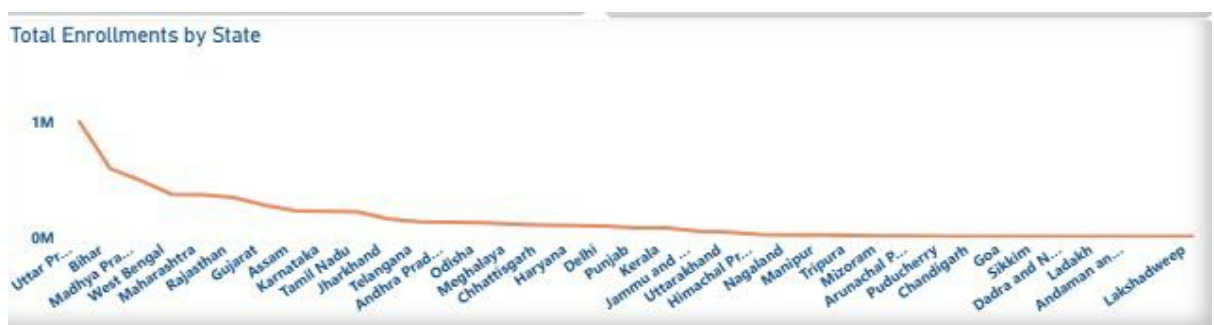




### → Infrastructure vs Awareness Gap :

**Rajasthan**, despite being geographically large, shows comparatively **lower enrolment levels**, indicating possible **awareness or accessibility gaps**, whereas **Assam**, a smaller state, demonstrate **higher enrolment intensity**, reflecting better infrastructure outreach.

*Recommendation* : Targeted **awareness campaigns** in Rajasthan and other states with lower enrolments, and replication of **best practices** in high enrolments states.



## **Conclusion :**

This project demonstrates how anonymised Aadhar enrolment data can be transformed into actionable insights through structured analysis and visualisation. The findings can support informed decision-making for enrolment planning, infrastructure deployment, and service optimisation at the state level.